

Medial-axis reconstruction on complete Riemannian manifolds: capture, drift, and the unconditional theorem*

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Abstract

For a closed set X in a complete Riemannian manifold M , the *two-geodesic medial axis* M_X^g is the set of points admitting two distinct minimising geodesics to X (the cut locus of X), and the reconstruction $\mathcal{R}_X^g = \bigcup_{a \in M_X^g} B(a, d_X(a))$ is the union of its open medial balls. Across the series, the generalisation of Białożyty's Theorem 6 to Riemannian manifolds had been proved up to two flagged gaps: the total-reconstruction theorem on compact manifolds rested on an unproven extension of Lieutier's swallow property past cut points [5, Rem. 8.4], [6, Rem. 6.5], and the Hadamard characterisation for non-compact contact sets carried a radial-offset caveat [4, Thm. 3.4]. This note closes both, with no genericity, analyticity, curvature, or spread hypothesis. The first input is a *capture theorem*: along any solution of the differential inclusion $\Phi' \in -\text{conv } U(\Phi)$, where U is the (upper semicontinuous, in general non-closed-graph-selectable) field of unit directions of minimising geodesics to X , the distance d_X grows at unit rate for exactly as long as the trajectory avoids M_X^g ; hence every point of $M \setminus X$ is either *captured* — it lies at depth $\geq d_X(p)$ inside a medial ball of M_X^g — or its trajectory *escapes* to infinity, which is impossible when M is compact. Consequently $\mathcal{R}_X^g = M \setminus X$ for *every* compact M and every closed $X \neq \emptyset$ (even a single point). The subtlety that defeated the margin arguments — M_X^g is not closed, and the margin is not monotone through multi-geodesic points — is dissolved by stopping at the *first approach* to M_X^g rather than flowing through it; focal and conjugate degenerations are ordinary points of the inclusion. The second input is a horofunction formulation: calling ψ a *medial limit function* if $\psi = \lim [d(\cdot, a_k) - d_X(a_k)]$ locally uniformly with $a_k \in M_X^g$, the swallow lemma becomes three lines — $\{\psi < 0\} \subseteq \mathcal{R}_X^g$ — with no anchoring or pinching, and

$$\mathcal{R}_X^g = \bigcup \{ \{ \psi < 0 \} : \psi \text{ a medial limit function} \}$$

holds on *every* complete M . On a Hadamard manifold ideal medial limit functions are Busemann functions up to level, which resolves the offset caveat of [4]: a drift of offset c is exactly a medial limit at level c , reconstructing $\{b_\xi < -c\}$ — a statement, no longer a gap. Together with the trichotomy captured/drift-swallowed/escaping, this completes the programme: the

*Final note of a series generalising Theorem 6 of [1]: [2] carried its sufficiency half to Hadamard manifolds, [3] proved the full characterisation in real hyperbolic space, [4] gave the faithful curvature-free characterisation for every Hadamard manifold, [5] crossed into positive curvature, and [6] organised the theory into its two axes (compactness; the Busemann–Hessian sign). This note removes the last analytic caveats and states the theorem on an arbitrary complete Riemannian manifold. Drafted with the assistance of Claude (Anthropic); all arguments were re-derived and checked by hand, and the two mechanics of the capture theorem — the rate-one identity off the axis and the margin at first approach — were verified numerically on $\mathbb{S}^2$ against a non-circular convex contact curve, whose focal region forces the foot to slide (60 random starts: growth rate 1 to five digits, margin deficit $< 3 \cdot 10^{-6}$; file `medial_capture_check.py`). Remaining errors are ours.

Euclidean, Hadamard, spherical, spread, and compact theorems are all specialisations, and the remaining open problems are exactly the faithful (M_X^g -free) certificates on the escape side.

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1 Introduction

For a closed set $X \neq \emptyset$ in a complete Riemannian manifold M write $d_X = \text{dist}(\cdot, X)$,

$$M_X = \{a : a \text{ has } \geq 2 \text{ nearest points in } X\}, \quad M_X^g = \{a : a \text{ has } \geq 2 \text{ minimising geodesics to } X\},$$

for the two-*point* and two-*geodesic* medial axes ($M_X \subseteq M_X^g$; the endpoints of the two geodesics may coincide), and

$$\mathcal{R}_X = \bigcup_{a \in M_X} B(a, d_X(a)), \quad \mathcal{R}_X^g = \bigcup_{a \in M_X^g} B(a, d_X(a))$$

for the corresponding reconstructions by open medial balls. Białożyty’s Theorem 6 [1] characterises $\mathcal{R}_X$ in $\mathbb{R}^n$ (where $M_X = M_X^g$); the companion notes carried the characterisation to real hyperbolic space [3], to every Hadamard manifold with a faithful horoconvexity certificate [2, 4], and into positive curvature [5], and [6] organised the results into two independent axes: *compactness* governs partial-versus-total, the *Busemann–Hessian sign* governs the detection flavour.

Two analytic gaps remained, both flagged explicitly in the series.

- (G1) *The containment step on compact manifolds.* The total-reconstruction theorem $\mathcal{R}_X^g = M \setminus X$ for compact M [5, Thm. 8.3], [6, Thm. 6.4] used the ascending generalised-gradient flow of d_X and needed the starting point to remain inside the moving free ball. The elementary constant-margin identity proves this only while the reach stays below the cut distance of the current foot; past it, the argument invoked “Lieutier’s swallow property extending past cut points — expected, and unconditional for generic or analytic metrics, but not delivered by the margin identity alone” [5, Rem. 8.4], [6, Rem. 6.5].
- (G2) *The radial offset for non-compact contact sets.* In the Hadamard characterisation for arbitrary closed X [4, Thm. 3.4], a purely asymptotic drift in an inhomogeneous manifold could carry a nonvanishing radial offset, and then only the deep part of the defective horoball was proved reconstructed.

This note closes (G1) and (G2) and states the resulting theorem in full generality: *every* complete Riemannian manifold, *every* closed set. No curvature bound, genericity, analyticity, spread, or cardinality hypothesis is used anywhere.

The two inputs

Capture (§3). The obstruction in (G1) was never the existence of the ascending flow; it was the bookkeeping of the margin through multi-geodesic points, where the flow turns and the margin genuinely need not be monotone (Remark 3.4). The repair is to reverse the roles: follow *any* trajectory of the differential inclusion driven by the minimising-geodesic directions, and *stop at the first approach to M_X^g* . Off M_X^g the direction field is single-valued, the trajectory is a unit-speed curve along which d_X grows at rate exactly one — through focal and conjugate points without comment — so

$$d_X(\Phi_t) = d_X(p) + t, \quad d(p, \Phi_t) \leq t,$$

and the starting point sits at depth $\geq d_X(p)$ in every moving ball. The trajectory therefore either approaches M_X^g in finite time — *capture*: some $a \in M_X^g$ has $d(p, a) < d_X(a)$, i.e. $p \in \mathcal{R}_X^g$ — or avoids M_X^g forever and $d_X \rightarrow \infty$ along it — *escape*, impossible on a compact manifold. The fact that M_X^g is not closed (minimising geodesics can merge in the limit: focal degenerations) is harmless, because approach suffices: the margin is continuous. This is Theorem 3.1, and it makes the compact theorem unconditional (Corollary 3.2), for every closed $X \neq \emptyset$ — including $|X| = 1$, which the two-point axis cannot see at all.

Drift (§4). Capture cannot be the whole story on a non-compact manifold: a point can be reconstructed by medial balls whose centres run to infinity while its own trajectory escapes (the open half-planes of two points in $\mathbb{R}^2$ are the primal example). The right limit object costs nothing in a complete manifold: M is proper (Hopf–Rinow), so the functions $\psi_a = d(\cdot, a) - d_X(a)$ — the *free-ball functions* of medial centres, normalised by their own reach, with no basepoint — are precompact in the topology of locally uniform convergence along any sequence that reconstructs a fixed point. Calling any such limit a *medial limit function*, the swallow lemma of [4] becomes three curvature-free lines with no anchoring hypothesis: $\{\psi < 0\} \subseteq \mathcal{R}_X^g$ for every medial limit function ψ (Lemma 4.1). The characterisation on an arbitrary complete manifold follows (Theorem 4.3), together with the trichotomy: every $p \in M \setminus X$ is captured, drift-swallowed, or escaping-unswallowed, and $\mathcal{R}_X^g$ is exactly the first two classes (Theorem 4.4). On a Hadamard manifold the ideal medial limit functions are Busemann functions up to an additive level, and the offset caveat (G2) dissolves into a definition: a drift with asymptotic offset c is a medial limit function at level c and reconstructs $\{b_\xi < -c\}$ in full (Remark 4.6).

What the completed theory says

Section 5 re-derives every earlier result as a specialisation and updates the 2×2 of [6]: the compact row is unconditionally total for $\mathcal{R}_X^g$; the non-compact row is governed by the medial limit functions, whose faithful (M_X^g -free) certificates are known exactly on the geometries where the series established them — conv in $\mathbb{R}^n$ [1], hconv on Hadamard manifolds [4], vacuous on compact M — and open in mixed curvature. Those certificates, and the quantitative cut-shielding threshold for the two-point axis [6, Rem. 6.6], are the honest residue of the programme (§6).

2 Preliminaries

Throughout, M is a complete Riemannian manifold (hence a proper geodesic metric space, by Hopf–Rinow), $X \subseteq M$ is closed with $\emptyset \neq X \neq M$, and $d_X = \text{dist}(\cdot, X)$. All geodesics are unit-speed. For $x \in M \setminus X$ let

$$U(x) = \{\gamma'(0) : \gamma : [0, d_X(x)] \rightarrow M \text{ minimising geodesic, } \gamma(0) = x, \gamma(d_X(x)) \in X\} \subseteq T_x^1 M,$$

the set of unit initial directions of minimising geodesics to X . Since a geodesic is determined by its initial vector, distinct minimising geodesics have distinct initial directions, so

$$M_X^g = \{x \in M \setminus X : |U(x)| \geq 2\}.$$

We write $\Sigma := M_X^g$ when we use it as a stopping set, and $r(a) = d_X(a)$ for the reach of a centre.

Lemma 2.1 (Direction field: compactness and semicontinuity). *For every $x \in M \setminus X$ the set $U(x)$ is nonempty and compact, and the multimap $x \mapsto U(x)$ is upper semicontinuous: if $x_k \rightarrow x$ in $M \setminus X$ and $u_k \in U(x_k)$, $u_k \rightarrow u$ (in TM), then $u \in U(x)$. In particular U is single-valued and continuous at every point of $(M \setminus X) \setminus \Sigma$.*

Proof. Nonemptiness and compactness: minimisers exist by properness, and a limit of initial vectors of minimising geodesics from the same x is the initial vector of a limit geodesic, which minimises by continuity of d_X and of lengths. Upper semicontinuity: let γ_k be the minimising geodesics with $\gamma_k'(0) = u_k$. By smooth dependence of geodesics on initial conditions, $\gamma_k \rightarrow \gamma$ locally uniformly where $\gamma'(0) = u$; the lengths $d_X(x_k) \rightarrow d_X(x)$, and the endpoints $\gamma_k(d_X(x_k)) \in X$ converge to $\gamma(d_X(x)) \in X$ (X closed). So γ is a geodesic from x to X of length $d_X(x)$: minimising, i.e. $u \in U(x)$. At a point with $U(x) = \{u\}$, upper semicontinuity of a compact-valued multimap into a singleton forces continuity of every selection near x — note this is continuity at x only; U may be multivalued arbitrarily close by. $\square$

Lemma 2.2 (First variation of d_X ; [12, §3]). *For $x \in M \setminus X$ and $w \in T_x M$, the one-sided directional derivative of d_X exists and equals*

$$\left. \frac{d}{ds} \right|_{s=0^+} d_X(\exp_x(sw)) = \min_{u \in U(x)} \langle w, -u \rangle.$$

Proof sketch. “ $\leq$ ”: fix $u \in U(x)$ with foot $y = \gamma(d_X(x))$ and vary γ through curves from $\exp_x(sw)$ to y ; the first variation of arc length gives $d_X(\exp_x(sw)) \leq d(\exp_x(sw), y) \leq d_X(x) + s \langle w, -u \rangle + O(s^2)$. “ $\geq$ ”: for $s_k \downarrow 0$ pick $u_k \in U(\exp_x(s_k w))$; by Lemma 2.1 a subsequence converges to some $u \in U(x)$, and the reverse comparison along the geodesics with directions u_k gives the matching lower bound with this u , hence with the minimum. Details (with the semiconcavity of d_X on $M \setminus X$, which we do not need) are in Mantegazza–Mennucci [12]. $\square$

The descent inclusion. On the open set $M \setminus X$ define the multimap

$$G(x) := \text{conv}(-U(x)) \subseteq T_x M.$$

By Lemma 2.1, G is upper semicontinuous with nonempty compact convex values contained in the closed unit ball. A *capture curve* from $p \in M \setminus X$ is a locally Lipschitz curve $\Phi : [0, T_{\max}) \rightarrow M \setminus X$ with $\Phi_0 = p$ and

$$\Phi'(t) \in G(\Phi_t) \quad \text{for a.e. } t, \tag{1}$$

maximally extended in $M \setminus X$. Local existence of solutions of (1), and extension as long as the trajectory stays in a compact subset of the open domain, are the standard theory of differential inclusions with upper semicontinuous convex compact right-hand side (Aubin–Cellina [7, §2.1]), applied in charts and concatenated; $|\Phi'| \leq 1$ a.e., so trajectories are 1-Lipschitz. We emphasise that *no* uniqueness, canonicity, or semiconcave-gradient selection is needed: any solution serves. (At a point of Σ the inclusion may branch; off Σ it is the classical ODE of the field $-u$.) When U is single-valued along a trajectory the curve simply extends the current minimising geodesic beyond x , switching fields continuously as the foot slides.

Free-ball functions. For $a \in M \setminus X$ set

$$\psi_a := d(\cdot, a) - d_X(a) : M \rightarrow \mathbb{R},$$

a 1-Lipschitz function, negative exactly on the open free ball $B(a, d_X(a))$, with $\psi_a \geq 0$ on X . It is normalised by the reach — $\min \psi_a = -d_X(a)$ at a — rather than by a basepoint; this intrinsic normalisation is what makes the limits below basepoint-free.

Definition 2.3 (Medial limit functions). *A function $\psi : M \rightarrow \mathbb{R}$ is a medial limit function of X if there are $a_k \in M_X^g$ with $\psi_{a_k} \rightarrow \psi$ locally uniformly. It is finite if (a_k) has a bounded subsequence and ideal otherwise (then ψ is a horofunction-type limit: $d(a_k, \cdot) \rightarrow \infty$ locally uniformly while ψ stays finite). Every ψ is 1-Lipschitz, and $\psi \geq 0$ on X (each ψ_{a_k} is), so $\{\psi < 0\}$ is an open set disjoint from X : the free horoball of ψ . For finite ψ along $a_k \rightarrow a$: if $a \in M_X^g$ then $\psi = \psi_a$ is a free-ball function; in general (M_X^g is not closed) $\psi = \psi_a$ with a a focal-degenerate limit of medial centres — still usable, by Lemma 4.1, which never needs the limit centre itself.*

Remark 2.4 (Relation to the horofunction compactification). M proper makes $C(M)$ with the topology of locally uniform convergence a countably compact home for 1-Lipschitz functions normalised at a point (Arzelà–Ascoli). The classical horofunction compactification [10, 9] normalises $d(\cdot, a_k)$ by a basepoint value; our ψ ’s differ from it by the scalar $d(o, a_k) - d_X(a_k)$, which is exactly the *offset* bookkeeping of [4, Thm. 3.4]. Working with reach-normalised limits absorbs the offset into the function itself; see Remark 4.6. On a Hadamard manifold every ideal medial limit function is a Busemann function plus a constant (horofunctions of a proper CAT(0) space are Busemann [9, II.8.13]), so ideal medial limit functions are exactly “Busemann function b_ξ + level c ”.

3 The capture theorem

Theorem 3.1 (Capture or escape). *Let M be complete, X closed, $p \in M \setminus X$, and let Φ be any capture curve from p . Let*

$$T := \sup\{t \in [0, T_{\max}) : \Phi_s \notin \Sigma \ \forall s \in [0, t]\}.$$

Then on $[0, T)$,

$$d_X(\Phi_t) = d_X(p) + t, \quad d(p, \Phi_t) \leq t, \quad \text{hence} \quad d_X(\Phi_t) - d(p, \Phi_t) \geq d_X(p), \quad (2)$$

and exactly one of the following holds.

- (i) (Capture.) $T < T_{\max}$. Then there exist $a \in M_X^g$ arbitrarily close to Φ_T ($a = \Phi_T$ itself if $\Phi_T \in \Sigma$) with

$$d_X(a) - d(p, a) \geq d_X(p) - \varepsilon \quad (\varepsilon > 0 \text{ arbitrary}),$$

so $p \in B(a, d_X(a))$ for such a : $p \in \mathcal{R}_X^g$, at depth arbitrarily close to $d_X(p)$, and $T \leq \text{diam}(M) - d_X(p)$.

- (ii) (Escape.) $T = T_{\max}$. Then $T_{\max} = \infty$, the trajectory never meets Σ , $d_X(\Phi_t) = d_X(p) + t \rightarrow \infty$, and Φ_t leaves every compact subset of M . This is possible only if M is non-compact.

Proof. The rate-one identity. Fix $t \in [0, T]$; on $[0, t]$ the trajectory avoids Σ , so $U(\Phi_s) = \{u(\Phi_s)\}$ is a singleton for every $s \leq t$ and (1) reads $\Phi'(s) = -u(\Phi_s)$ at a.e. s . The map $h(s) := d_X(\Phi_s)$ is Lipschitz. Let s be any point where both $\Phi'(s)$ exists (with the inclusion valid) and $h'(s)$ exists — almost every $s \in [0, t]$. Write $v = \Phi'(s) = -u(\Phi_s)$, a unit vector. In a chart around Φ_s , differentiability gives $d(\Phi_{s+\sigma}, \exp_{\Phi_s}(\sigma v)) = o(\sigma)$ as $\sigma \downarrow 0$, and d_X is 1-Lipschitz, so

$$h(s + \sigma) = d_X(\exp_{\Phi_s}(\sigma v)) + o(\sigma) = h(s) + \sigma \min_{u \in U(\Phi_s)} \langle v, -u \rangle + o(\sigma) = h(s) + \sigma \langle -u, -u \rangle + o(\sigma),$$

by Lemma 2.2 and single-valuedness. Hence $h'(s) = |u|^2 = 1$ at a.e. $s \in [0, t]$, and integrating, $h(t) = d_X(p) + t$. The bound $d(p, \Phi_t) \leq t$ is the 1-Lipschitz property of the trajectory, and the margin inequality in (2) is their difference. Note that no closedness of Σ , and no structure at focal or conjugate points, was used: a point of $(M \setminus X) \setminus \Sigma$ in the closure of Σ is simply a point where U is single-valued, and the identity passes through it.

Dichotomy. Suppose first $T < T_{\max}$. If $\Phi_T \in \Sigma$, continuity of $d_X(\Phi) - d(p, \Phi)$ and (2) on $[0, T]$ give $d_X(\Phi_T) - d(p, \Phi_T) \geq d_X(p) > 0$, so $a = \Phi_T \in M_X^g$ contains p in its open medial ball at depth $\geq d_X(p)$. If $\Phi_T \notin \Sigma$, then by definition of T as a supremum, every interval $(T, T + \delta] \subseteq [0, T_{\max})$ contains a time s_δ with $\Phi_{s_\delta} \in \Sigma$; letting $\delta \downarrow 0$ and using continuity of the margin, $a = \Phi_{s_\delta} \in M_X^g$ satisfies $d_X(a) - d(p, a) \geq d_X(p) - \varepsilon$ for δ small. In both cases $p \in B(a, d_X(a))$ with $a \in M_X^g$, i.e. $p \in \mathcal{R}_X^g$; and $d_X \leq \text{diam}(M)$ bounds T via (2).

Now suppose $T = T_{\max}$: the trajectory never meets Σ , and (2) holds on all of $[0, T_{\max})$. If $T_{\max} < \infty$, then $\Phi([0, T_{\max})) \subseteq \bar{B}(p, T_{\max}) \cap \{d_X \geq d_X(p)\}$, a compact subset of the open set $M \setminus X$ (properness of M); by the extension property of solutions this contradicts maximality. So $T_{\max} = \infty$ and $d_X(\Phi_t) = d_X(p) + t \rightarrow \infty$; since d_X is bounded on compact sets, Φ_t leaves every compact set. On compact M , $d_X \leq \text{diam}(M)$ makes this impossible, so case (i) occurs. $\square$

Corollary 3.2 (Total reconstruction on compact manifolds, unconditional). *Let M be a compact Riemannian manifold and $X \subseteq M$ closed, $\emptyset \neq X \neq M$. Then*

$$\boxed{\mathcal{R}_X^g = M \setminus X},$$

and quantitatively: every $p \in M \setminus X$ lies at depth $\geq d_X(p) - \varepsilon$ (any $\varepsilon > 0$) inside a medial ball of M_X^g . No curvature, genericity, analyticity, spread, or cardinality hypothesis is required; in particular $|X| = 1$ is allowed, where the two-point axis has $M_X = \emptyset$ but the two-geodesic axis still reconstructs everything. This closes the containment caveat of [5, Rem. 8.4] and [6, Rem. 6.5]: Theorem 8.3 of [5] and Theorem 6.4 of [6] hold as stated, with “resting on Lieutier’s property” deleted.

Proof. $\mathcal{R}_X^g \subseteq M \setminus X$ since medial balls are free. Conversely, if $p \in M_X^g$ then $p \in B(p, d_X(p)) \subseteq \mathcal{R}_X^g$; otherwise start a capture curve at p (local existence, §2); by Theorem 3.1 escape is impossible on compact M , so p is captured. $\square$

Corollary 3.3 (Two-point axis under spread; cf. [5, Thm. 6.1]). *If M is compact and $d_X < \text{inj}(M)$ on $M \setminus X$ — e.g. if X is δ -dense with $\delta < \text{inj}(M)$ — then $\mathcal{R}_X = M \setminus X$ for the two-point axis as well.*

Proof. By Corollary 3.2 every p lies in $B(a, d_X(a))$ for some $a \in M_X^g$. Two distinct minimising geodesics from a with the same endpoint x_0 would put x_0 in the cut locus of a , giving $d_X(a) = d(a, x_0) \geq \text{inj}(M)$, contradiction. So the two geodesics of a have distinct endpoints: $a \in M_X$. $\square$

Remark 3.4 (Why stopping at Σ , not flowing through it, is the correct formulation). Two natural strengthenings of Theorem 3.1 are false, and identifying their failure is what shaped the proof. First, *the margin $d_X(\Phi) - d(p, \Phi)$ is not monotone through multi-geodesic arcs*, already in $\mathbb{R}^2$: for $X = \{(0, 0), (0, 2)\}$ and $p = (1, 0.9)$, the Lieutier flow reaches the bisector $\{y = 1\}$ and slides along it, and the margin decreases strictly (from ≈ 1.345 toward the limit 1, the depth of p in the limiting half-plane). Second, even the quadratic margin $d_X^2 - d(p, \cdot)^2$, which *is* monotone along unique-foot and bisector arcs in $\mathbb{R}^n$, fails at turning points where the active contact set changes (a three-point configuration in $\mathbb{R}^2$ suffices: $X = \{(0, 0), (0, 2), (10, 5)\}$, with the flow from $p = (1, 0.9)$ turning at $\approx (5.75, 1)$, where the derivative of the quadratic margin is negative). Both phenomena live *on* Σ ; on its complement the rate-one identity (2) is exact. Since reaching Σ is the goal, not an obstacle, the correct move is to stop there — capture is achieved, with the margin intact, *before* the first multi-geodesic time. The failure of closedness of Σ (focal degenerations, e.g. the evolute of an ellipse in $\mathbb{R}^2$, or the conjugate-cusp endpoints of an ellipsoid’s cut segment) is equally harmless: such points are unique-geodesic points, the field $-u$ is continuous at them (Lemma 2.1), and the trajectory passes through with d_X still growing at rate one, the foot sliding continuously.

Remark 3.5 (Relation to Lieutier’s flow). For semiconcave d_X one can canonically select the minimal-norm element of G (Lieutier’s generalised gradient [11], in $\mathbb{R}^n$; [8, 12] for the semi-concavity). Our argument uses the opposite regime: it only ever analyses the inclusion *off* Σ , where every selection agrees with the naive one (extend the unique minimising geodesic), and it terminates at Σ , where Lieutier’s flow is designed to continue. This is why no analogue of the swallow property past cut points is needed: the trajectory is stopped at first approach, and the reconstruction is read off the margin accumulated strictly before it. The price — capture produces a centre only *near* the first Σ -approach when $\Phi_T \notin \Sigma$ — costs an arbitrarily small ε of depth, invisible to the open-ball statement.

4 Drift: the swallow, the characterisation, the trichotomy

Capture is complete on compact manifolds, but on a non-compact M a point can be reconstructed by medial balls whose centres go to infinity while its own capture curve escapes. The primal example is $X = \{x_0, x_1\} \subseteq \mathbb{R}^2$: a point p in an open half-plane bounded by the line x_0x_1 , placed beyond x_0 at an obtuse angle, has a capture curve that runs straight to infinity with the single foot x_0 — yet p is swallowed by the bisector balls $B(a_k, r(a_k))$, $a_k \rightarrow \infty$. The limit of those balls is a half-plane, and the correct general home for such limits is the space of medial limit functions of Definition 2.3.

Lemma 4.1 (Swallow, general and offset-free). *Let M be any complete Riemannian manifold, X closed, and ψ a medial limit function of X . Then*

$$\{\psi < 0\} \subseteq \mathcal{R}_X^g.$$

If the witnessing centres lie in M_X , then $\{\psi < 0\} \subseteq \mathcal{R}_X$.

Proof. Let $\psi = \lim \psi_{a_k}$ locally uniformly, $a_k \in M_X^g$, and let $q \in \{\psi < 0\}$. Then $\psi_{a_k}(q) \rightarrow \psi(q) < 0$, so for large k

$$d(q, a_k) - d_X(a_k) = \psi_{a_k}(q) < 0,$$

i.e. $q \in B(a_k, d_X(a_k))$ with $a_k \in M_X^g$: $q \in \mathcal{R}_X^g$. $\square$

No anchoring of feet, no curvature bound, no pinching, no offset hypothesis: the entire content of the swallow lemmas of [4, Lem. 2.2] (and its constant-margin spherical form, [5, Rem. 4.2]) is absorbed into the normalisation of ψ_a by the reach.

Lemma 4.2 (Compactness of reconstructing sequences). *Let $p \in \mathcal{R}_X^g$, and let $a_k \in M_X^g$ be any sequence with $\limsup_k [d_X(a_k) - d(p, a_k)] > 0$ (e.g. a sequence of centres whose balls contain p with margin bounded away from 0; a single ball repeated qualifies). Then a subsequence of (ψ_{a_k}) converges locally uniformly to a medial limit function ψ with $\psi(p) < 0$.*

Proof. The functions ψ_{a_k} are 1-Lipschitz and $\psi_{a_k}(p) = d(p, a_k) - d_X(a_k)$ is bounded: below by $-d_X(p)$ (since $d_X(a_k) \leq d_X(p) + d(p, a_k)$) and above by 0 along the qualifying subsequence. By Arzelà–Ascoli on the proper space M , a further subsequence converges locally uniformly; the limit is a medial limit function, and $\psi(p) = \lim \psi_{a_k}(p) \leq -\limsup [d_X(a_k) - d(p, a_k)] < 0$. $\square$

Theorem 4.3 (Characterisation on an arbitrary complete manifold). *For every complete Riemannian manifold M and every closed $X \subseteq M$, $\emptyset \neq X \neq M$,*

$$\mathcal{R}_X^g = \bigcup \{ \{\psi < 0\} : \psi \text{ a medial limit function of } X \}.$$

The same identity holds for $\mathcal{R}_X$ and M_X in place of $\mathcal{R}_X^g$ and M_X^g . (Every finite medial limit function is itself a free-ball function ψ_a , with a in the closure of M_X^g in $M \setminus X$; the sublevel set of such a limit is still swallowed, by the lemma, even when $a \notin M_X^g$.)

Proof. “ $\supseteq$ ” is Lemma 4.1. “ $\subseteq$ ”: if $p \in \mathcal{R}_X^g$ then $p \in B(a, d_X(a))$ for some $a \in M_X^g$, and ψ_a — a medial limit function via the constant sequence — has $\psi_a(p) < 0$. $\square$

The identity is deliberately economical: its force is not the set equality (the finite members alone give a tautology) but the pair of facts it packages. *First*, the ideal members reconstruct their entire open sublevel set — the statement that in [3, 4] required anchoring, pinching, or bisector witnesses, and now holds bare. *Second*, combined with Theorem 3.1 it yields the trichotomy that generalises the hull/drift/ray-escape structure of [4, Thm. 2.1] to arbitrary complete manifolds.

Theorem 4.4 (Trichotomy on a complete manifold). *Let M be complete, X closed, $p \in M \setminus X$. Exactly one of the following holds.*

- (1) (Captured.) *Some capture curve from p approaches M_X^g in finite time. Then $p \in \mathcal{R}_X^g$, at depth arbitrarily close to $d_X(p)$.*
- (2) (Swallowed, uncaptured.) *No capture curve from p approaches M_X^g in finite time, but $\psi(p) < 0$ for some medial limit function ψ . Then $p \in \mathcal{R}_X^g$: reconstructed, but only by centres p ’s own ascent cannot find — a remote ball, or a drifting sequence.*
- (3) (Escaping.) *Neither. Then $p \notin \mathcal{R}_X^g$, and every capture curve from p escapes to infinity with $d_X(\Phi_t) = d_X(p) + t$.*

In particular $\mathcal{R}_X^g = \{p : (1) \text{ or } (2)\}$, and on compact M class (1) is all of $M \setminus X$.

Proof. The classes are exclusive and exhaustive by construction. (1) $\Rightarrow p \in \mathcal{R}_X^g$ is Theorem 3.1(i); (2) $\Rightarrow p \in \mathcal{R}_X^g$ is Lemma 4.1. Conversely, if $p \in \mathcal{R}_X^g$ then $p \in B(a, d_X(a))$ for some $a \in M_X^g$ and the finite medial limit function ψ_a has $\psi_a(p) < 0$, so p lies in class (2) whenever it is not in class (1) — never in class (3). Finally, a point not in class (1) has every capture curve uncaptured, hence escaping, by the dichotomy of Theorem 3.1. $\square$

Remark 4.5 (The witnesses of class (2)). Both kinds of witness genuinely occur, already for $X = \{x_0, x_1\} \subseteq \mathbb{R}^2$ and p beyond x_0 in an open half-plane, at an obtuse angle to the bisector: every capture curve from p runs straight to infinity with the single foot x_0 , yet p lies in a single *finite* bisector ball (and also in the ideal half-plane limit of the bisector balls). Class (2) is thus “reconstruction is non-local for p ”: the certificate cannot be found by ascending from p , which is exactly the phenomenon that separates the non-compact theory from the compact one and forces the faithful certificates of [1, 3, 4] to quantify over supporting half-spaces or horoballs rather than over flow lines.

Remark 4.6 (The offset caveat of [4], resolved). Let M be Hadamard. Ideal medial limit functions are exactly $b_\xi + c$ with $\xi \in \partial_\infty M$ and $c \in \mathbb{R}$ (Remark 2.4), normalised so that b_ξ vanishes on the horosphere of the maximal free horoball $\mathcal{H}$ at ξ ; necessarily $c \geq 0$ when X meets every neighbourhood of that horosphere. Call $\mathcal{H}$ *defective at level c* if $b_\xi + c$ is a medial limit function; then Lemma 4.1 reconstructs $\{b_\xi < -c\}$ — the horoball shrunk by c . The “radial offset” $\text{off}_k = d(o, a_k) + b_\xi(a_k)$ of [4, Thm. 3.4] is precisely the discrepancy $\psi_{a_k} - (b_\xi \circ \cdot)$ evaluated along the drift: a drifting sequence with $\text{off}_k \rightarrow c$ is a medial limit at level c . The single unresolved case of [4, Thm. 3.4] — a purely asymptotic drift with non-vanishing offset in an inhomogeneous manifold, where only the deep part $\{\tau_p > \limsup \text{off}_k\}$ was reconstructed — is therefore not a gap in the theorem but the exact content of the level: the drift *is* a level- c certificate and reconstructs exactly $\{b_\xi < -c\}$, while the full-horoball statement is the assertion that a level-0 certificate exists. Under any of the conditions of [4, Thm. 3.4] (pinching, flat case, a bisector witness on the horosphere — hence in every model of the series) level 0 is achieved and the horoball is reconstructed in full; the hconv-certificate of [4, Thm. 3.3–3.4] detects defectiveness at *some* level unconditionally. The characterisation of the exact level

$$c(\xi) := \inf\{c \geq 0 : b_\xi + c \text{ is a medial limit function}\}$$

from X alone, in an inhomogeneous manifold, is the sharpened form of the remaining question, and it is a question about X and the horosphere geometry, not about the medial axis.

5 Specialisations: the series re-derived

Euclidean space [1]. In $\mathbb{R}^n$ geodesics between two points are unique, so $M_X^g = M_X$ and $\mathcal{R}_X^g = \mathcal{R}_X$. Ideal medial limit functions are affine: $|y - a_k| - r_k$ with $a_k \rightarrow \infty$ subconverges to $y \mapsto -\langle y, u \rangle - c$, whose sublevel set is an open half-space. Theorem 4.3 thus reads: $\mathcal{R}_X$ is the union of the medial balls and of the open half-spaces that are limits of medial balls — and Theorem 6 of [1] is its faithful certificate: such a half-space limit exists iff a supporting half-space has non-convex contact set (Motzkin), and the level is then 0, the drift running over a bisector witness with no offset. The escaping class (3) is exactly Białożył’s unreconstructed set — the rays leaving X with a single foot and no defective facet around them.

Hadamard manifolds [2, 3, 4]. Here class (2)’s ideal witnesses are Busemann functions at a level (Remark 4.6); the faithful certificate for their existence is the horoconvexity defect $X \cap \partial\mathcal{H} \neq \text{hconv}(X \cap \partial\mathcal{H})$ of [4, Thm. 3.3–3.4], unconditional for the $\subseteq$ direction and achieving level 0 in every model of the series. The hull inclusion $\widehat{X} \setminus X \subseteq \mathcal{R}_X$ [2, Thm. 3.1] populates classes (1)–(2) inside the hull; the trichotomy of [4, Thm. 2.1] (hull/drift/ray-escape) is Theorem 4.4 read through the visual boundary.

The sphere, compact manifolds, space forms [5, 6]. Corollary 3.2 contains: total reconstruction on $\mathbb{S}^n$ ([5, Thm. 4.1], now with the $|X| \geq 2$ hypothesis needed only for the two-point axis — $\mathcal{R}_X^g = \mathbb{S}^n \setminus X$ holds even for a single point, reconstructed from the antipode); total reconstruction on every compact M ([5, Thm. 8.3], [6, Thm. 6.4], caveats deleted); and, new, total $\mathcal{R}_X^g$ on every spherical space form $\mathbb{S}^n/\Gamma$ for *every* closed X — the sharp statement left open in [5, Rem. 7.3], where only the δ -dense case was settled. For the two-point axis the compact theory is unchanged: Corollary 3.3 under spread, and cut-shielding as the exact obstruction in general, with the prolate-ellipsoid transition at eccentricity $c^* = 2.6 \pm 0.1$ [6, Rem. 6.6] — shielded points are precisely class-(1) points of $\mathcal{R}_X^g$ whose capturing centres lie in $M_X^g \setminus M_X$.

Non-compact positive curvature. On a paraboloid of revolution ($\text{sec} > 0$, one flattening end) with X a small circle about the vertex, the vertex is a finite medial centre reconstructing the inner disk (class (1)); every point outside X on the flaring side ascends its meridian forever with a single foot and no medial limit function reaches it: class (3), $\mathcal{R}_X^g \subsetneq M \setminus X$. With X two points, the bisector is unbounded and its escaping medial centres furnish genuinely *ideal* medial limit functions on a $\text{sec} > 0$ manifold — consistent with the trichotomy of [6], since their sublevel sets are unbounded *non-convex* free regions (Busemann functions are concave above zero; it is the maximal free *convex* regions that are bounded there, not the carriers). Partial-versus-total is decided by compactness, detection flavour by the Hessian sign, exactly as the 2×2 prescribes — now with both rows unconditional.

The updated table. Rows: compactness (axis A). Columns: Busemann–Hessian sign (axis B). Entries: the two-geodesic theory, with the two-point refinement in brackets.

	$\text{sec} \leq 0$	$\text{sec} \equiv 0$	$\text{sec} > 0$
non-compact	partial; hconv defect at level $c(\xi)$ [4]	partial; conv defect (Białytyt [1])	partial (paraboloid) finite + ideal centres
compact	<i>total</i> $\mathcal{R}_X^g$ [M_X : shielding]	<i>total</i> $\mathcal{R}_X^g$ [M_X : shielding]	<i>total</i> $\mathcal{R}_X^g$ [M_X : shielding; = $\mathbb{S}^n$ ok]

The compact row is Corollary 3.2 — unconditional, all curvatures, $|X| \geq 1$. The non-compact row is Theorem 4.3, with the faithful certificates known on the flat and Hadamard columns and open elsewhere (§6).

6 What remains

The theory is now unconditional in its dynamical form on every complete Riemannian manifold. What remains open is exactly the *faithful* layer — certificates decidable from X and the geometry of M , with M_X^g absent — and it lives entirely on the non-compact (escape) side.

- (1) *Certificates in mixed curvature.* For which ideal points (horofunctions) h of a general non-compact M is $h + c$ a medial limit function, read from the contact structure of X on $\{h = 0\}$? Known: conv/Motzkin in $\mathbb{R}^n$ [1, 13]; hconv on Hadamard manifolds, with the Carnot and Alexandrov frontiers of [4, §9–10]; vacuous on compact M . Unknown already for a complete surface with both signs of curvature.
- (2) *The level function.* On an inhomogeneous Hadamard manifold, characterise $c(\xi) = \inf\{c : b_\xi + c \text{ medial limit}\}$ from X alone (Remark 4.6); $c(\xi) = 0$ under pinching, in the flat case, and over any bisector witness, and no example with $c(\xi) > 0$ is known — deciding whether the deep-part phenomenon of [4, Thm. 3.4] actually occurs is the sharpest open question on the Hadamard side.
- (3) *The two-point axis in variable curvature.* Quantify cut-shielding: the prolate threshold $c^* = 2.6 \pm 0.1$ awaits an analytic determination, and more generally the class of compact M with $\mathcal{R}_X = M \setminus X$ for all sparse X (constant curvature is sufficient; bounded eccentricity conjecturally) [5, 6].
- (4) *Chebyshev problems.* The detection questions the series isolated on its frontiers: Motzkin in Carnot groups ($\mathbb{CH}^n$ horospheres) and in Alexandrov spaces of indefinite curvature (pinched horospheres) [4, §10]; and the domain (Lieutier) problem on $\mathbb{S}^n$, where the boundary’s positively curved intrinsic geometry becomes the detection object [6, §7].

Conclusion. Białożyty’s Theorem 6 generalises to every complete Riemannian manifold, in two layers. The *unconditional* layer is finished here: reconstruction by the two-geodesic axis is exactly the union of the sublevel sets of the medial limit functions (Theorem 4.3); every point is captured, swallowed, or escaping (Theorem 4.4); capture is total on compact manifolds (Corollary 3.2), with no hypotheses whatsoever beyond completeness and closedness. The *faithful* layer — replacing “is a medial limit” by a certificate on contact sets — is exact on the flat knife-edge (conv, [1]), on every Hadamard manifold (hconv, [4]), and trivial on compact manifolds; its extension into mixed curvature, together with the Chebyshev-set problems it generates, is the genuine residue. The gaps flagged in the series — the containment step past cut points and the radial offset — are closed, the first by stopping the ascent at its first approach to the axis instead of flowing through it, the second by normalising the limit objects by their own reach.

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